A Universe from Coulomb and Newton

From a Ripple in the Electric Field to the Cosmic Web

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Abstract

A non-technical companion to the article *Coulomb + Newton: Towards a Cosmology.* It follows a single chain upward: a faint ripple in the electric field makes charge, in units of exactly one, and in two sizes — which are the proton and the electron; Coulomb's law then builds nuclei, atoms and chemistry out of them; and Newton's law, acting on the energy all that matter carries, gathers it into the cosmic web. No Big Bang singularity, no inflation, no dark-matter particle. The price, stated plainly at the end, is that it explains neither the accelerating expansion nor the spinning of galaxies.

One kind of model, all the way up

This essay begins not with the universe but with a way of describing matter. In the picture called RealQM, an atom is not a wave in an abstract many-dimensional space but something far more ordinary: *charge spread out in space as a density*, one cloud per electron, each cloud occupying its own region, the regions meeting at free boundaries that settle wherever the energy is least. That is the same kind of description — densities, forces, boundaries — that engineers use for water, steel and air. And because it is the same kind, it does not have to be abandoned as one moves up in scale: from a nucleus to an atom, from an atom to a material, with nothing about the mathematics changing on the way.

This essay carries that ladder to its top rung by adding **gravity**. Gravity acts on the energy that matter carries, and it is far too weak to matter until matter is piled up in astronomical quantities — so adding it extends the very same description to the scale of galaxies. Nothing in the mathematics changes: one more field, one more force term, the same continuum picture.

Where that leads is cosmology, and cosmology is exactly the right place for it, because cosmology is the subject where the smallest scale meets the largest. Stars shine by nuclear fusion: something happening at a millionth of a billionth of a metre sets how brightly a star burns, how long it lives, and what it seeds its galaxy with. The abundances of the light elements are set by nuclear reactions yet are read off the sky. And the missing mass of galaxies, a problem posed at the largest scales, is chased at the smallest — in particle colliders and underground detectors. A description that runs continuously from nuclei to galaxies is not an ambition imposed on cosmology; it is what cosmology has always quietly required.

What replaces the bang

Every cosmology needs an origin for its *matter*, if not for time itself. The standard one is the Big Bang: a hot dense state, with a brief episode of inflation stretching microscopic ripples into the seeds of galaxies. This picture asks for something quieter. Suppose the electric potential — the field whose slopes push charges around — is not perfectly flat but very faintly **rippled**.

Where the ripples are finest, the field curves sharply, and it is curvature that makes charge. Each tiny crest becomes a unit of positive charge, each trough a unit of negative: protons and electrons, in exactly equal numbers, because every crest has its trough. The graininess of matter is the graininess of the ripples.

Something pleasing then happens to the arithmetic. One might expect the strength of the ripples to be a free knob, tunable to taste. It isn't. Demanding that each ripple carry exactly *one* charge — no halves, no doubles — fixes how strong it must be, once you have said how small it is. And how small it is turns out to be the same unanswered question as how big a proton is. So the whole small-scale story runs on a single unknown length, with everything else following.

And with the charges, the masses. A charge cloud's energy is set by how tightly it is squeezed — squeezing costs energy, like compressing a spring — and a thing's mass is simply its energy content. So two sizes of cloud mean **two masses, and only two**: the small tight positive cloud is the heavy one, the **proton**; the large loose negative one is the light one, the **electron**. That there are exactly two stable kinds of matter follows from there being exactly two sizes — no in-between mass exists, any more than an in-between charge does. What does *not* follow is the actual number: a proton is 1836 times an electron, and the simple size-and-energy rule misses this by a factor of about ten. The *count* of masses is a result here; their *values* are an open question.

One thing the ripple is *not* asked to do. It makes the matter, but it is not asked to say where that matter ends up in the large — which patch of sky is crowded and which is empty. That large-scale unevenness is taken here as given, an input the picture does not derive. It is honest to mark it as a debt rather than to dress it up.

The small world: Coulomb builds matter

Electric charge comes in two signs, and in this picture they come out *different in size.* The positive charge forms small, dense clumps — **protons**. The negative charge spreads into large, diffuse clouds — **electrons**. From just these, Coulomb's law builds the rest by simple geometry: electrons bound around protons make **atoms**; electrons shared between atoms make **molecules** and all of chemistry.

Remarkably, the same force reaches inside the nucleus. In this picture a nucleus is not held together by a separate "strong force." It is protons glued by *compact* electrons — the very same electrons, squeezed small — again by ordinary Coulomb attraction, only at a tiny scale where it is correspondingly strong. What looks like a new and powerful nuclear force is just electricity at close range. One force, two scales.

The large world: Newton gathers matter

Now the second force. The matter Coulomb has built carries *energy*, and energy is what gravitates (a word on that below). Newton's law acts on it in the way everyone knows: masses attract.

But there is a subtlety here, and it is the heart of the picture. Gravity does not respond to how much matter there is; it responds to how much matter there is *compared with the average.* This is less exotic than it sounds — it is forced. Spread matter evenly through an endless space and ask what its gravity does at some point, and the question has no answer: the pull depends on the order in which you add up the infinitely many contributions, and the field has no well-defined value at all. Newton's law simply does not work on a uniform infinite universe. The cure, which every version of Newtonian cosmology must use in some form, is to measure matter from a baseline, and the obvious baseline is the average. What this picture adds is only that it takes the cure seriously, as physics rather than bookkeeping, and follows where it leads. A region denser than average pulls its surroundings in. A region emptier than average does the opposite: it *pushes* them away. So even though every scrap of matter is ordinary and positive, the thing gravity actually listens to comes in **two signs** — surplus and shortfall. Surplus then gathers with surplus, and the matter drains out of the shortfalls.

That is the whole story of the large scale. The surpluses become the glowing **cosmic web** of filaments and clusters that we live in; the shortfalls become the great **voids** between them — and the voids are not merely leftover empty space, but actively push their surroundings outward and so grow. No negative matter is needed anywhere: nothing falls upward, no strange substance is added. Emptiness relative to the average is repulsion enough.

One thing this picture does *not* give, and it is only honest to say so at once: it does not explain the accelerating expansion that astronomers attribute to dark energy. Voids push locally, which is real, but it is a local effect, and ordinary cosmology has it too. Across the universe as a whole the surpluses and the shortfalls cancel exactly — that is what "average" means — so the expansion neither slows nor speeds up. It **coasts**. That is a definite claim, and a risky one, because it can be checked against exploding stars across the sky; we return to it at the end.

The web as unmixing

There is a familiar name for what this sorting really is. Shake oil and water together, let the jar stand, and the two do not stay mixed: they *separate*, oil gathering with oil and water with water, until thin skins divide them into a pattern of blobs and films. Physicists call this **phase separation**, and it is one of the best-understood pattern-forming processes in nature — the very same thing that happens when a hot alloy cools and its two metals unmix into a marbled web.

The universe of surplus and shortfall does exactly this. Surplus gathers with surplus, and matter drains from the shortfalls, until the two are separated into a marbled pattern of filaments threaded through great empty rooms. Seen this way the glowing cosmic web is not a special or mysterious shape at all: it is the ordinary shape that things take when each gathers with its own kind — the same web you would get from unmixing oil and water, only drawn by gravity across the whole sky.

The analogy has one limit worth marking, because it is easy to overdraw. Oil and water are symmetric: swap them and the picture looks the same. Surplus and shortfall are *not.* A filament can go on getting denser without limit, but a void can at most become completely empty — there is a floor to emptiness and no ceiling to crowding. So the two are not mirror images of one another, and one should not expect voids to be shaped like filaments turned inside out. What halts the clumping before everything collapses to points is the ordinary springiness of matter:

squeeze a gas and it heats, and past a certain stiffness the heating wins.

A word about "mass"

We have been saying gravity acts on *mass*, then on *energy*, as if they were the same. They are. Mass, looked at closely, is just a name for *energy content*: a thing's resistance to being pushed is set by how much energy it holds — that is the real meaning of $E = mc^2$, and it is measured directly (a single particle can be weighed by its energy; a nucleus weighs less than its parts by exactly the energy of their binding). So gravity acts on the energy of matter, which is what we usually call its mass. We need no mysterious extra substance called mass; energy suffices.

One further temptation should be resisted, honestly. It is tempting to say a particle's energy is fixed *entirely* by the size of its charge cloud — small and tight meaning high energy, like a stiff spring. Qualitatively that is surely right (the small proton is the heavy one). But the exact numbers do not line up: the proton is more compact than a simple size–energy rule would make it, by about a factor of ten. So we say only what is safe — energy and size are closely *related*, not exactly *identified* — and we do *not* claim that gravity is somehow secretly derived from electricity. The two forces run side by side; whether they are ultimately one is an open question, not a claim of this picture.

Three numbers

Strip the picture to essentials and it rests on remarkably little: *three pure numbers*, each doing one job. The first is the strength of electricity — what physicists call the fine-structure constant, about 1/137 — and it fixes how much bigger an atom is than its nucleus: about twenty thousand times. The second is how much heavier a proton is than an electron: 1836. Those two, between them, settle all of atoms, chemistry and nuclei, and neither of them mentions gravity at all.

The third is the strength of gravity, and it is the strangest number in physics: measured against electricity, it is smaller by roughly 10^{39}. Pile up matter and gravity is negligible — until you have gathered something like 10^{57} protons, which is to say about the mass of the Sun, and then it wins. That single number is what separates the small world from the large one. And its absurd smallness is not an embarrassment: it is exactly why there is room for chemistry, for planets, for everything between an atom and a star. Were gravity anywhere near as strong as electricity, there would be no in-between at all, and nothing here to describe.

Compare the bookkeeping. The standard model of cosmology needs a singularity, an inflaton field, a dark-matter particle and a cosmological constant — a lengthening list of unseen ingredients. This picture asks for three numbers. One qualification, since it is easy to overstate: a term of the same kind as the cosmological constant does appear here, in the rule that only departures from the average gravitate. The difference is that it is not free to be chosen — it is fixed by the requirement that the universe's gravity cancels overall, and it thins out as the universe expands instead of staying fixed. The open questions are correspondingly few and sharp: none of the three is derived from anything deeper (nor are they in the standard account), and whether the two forces are at bottom one remains genuinely open. Three clean unknowns, in place of a menagerie of invisible substances — though, as the next section admits, the price is paid elsewhere.

A comparison with the standard model, and a soft spot in it

The standard cosmology — ΛCDM — rests on a hot Big Bang, with the cosmic microwave background (CMB) as its relic glow, supplemented by dark matter, dark energy and an early burst of inflation. Its single proudest piece of evidence is the CMB's spectrum: an almost perfect *Planck* (blackbody) curve, read as the thermal radiation of a hot, dense early universe once in equilibrium.

But this reads the evidence backwards. A perfect Planck spectrum is not something a hot dense gas naturally makes. In the author's analysis of blackbody radiation, a true Planck curve is the signature of a *resonant lattice* — a regular array of oscillators (a metallic crystal, a *gitter*) with a sharp high-frequency cutoff — and *not* of the disordered collisions of a hot gas or plasma, which have no such lattice and no such cutoff. On that analysis a hot dense fireball simply *cannot* generate a perfect Planck spectrum. (This contests the textbook view, by which any body in thermal equilibrium is supposed to radiate Planckian whatever its make-up; it is the author's position, developed at length in the blackbody work cited below.)

If so, the conclusion is sharp: **the CMB's near-perfect Planck spectrum is exactly what a hot dense gas would *not* produce** — so it cannot serve as evidence that any such hot dense initial state ever existed. **The CMB can say nothing about the existence of a hot dense beginning.** Its Planck form points, if anything, toward a resonant, lattice-like source — whatever and wherever that turns out to be — not toward a primordial fireball.

This is where the present picture simply steps aside. It has *no* hot Big Bang and no primordial fireball, so it neither needs nor claims the CMB as the relic of one. What the CMB is, and why it is so smooth and so Planckian, is left here as a separate open question — but its spectrum is *not* read as proof of a hot dense beginning, because on the blackbody analysis above it is no such proof.

And now the thing worth noticing

Only now, with both halves of the picture built, is it worth pointing out something that has not been used anywhere in building them. Coulomb's law and Newton's law have *the same shape*. Both fall off as one over distance squared; both are the same field equation. They differ in exactly two ways:

- **a sign.** For electric charge, *like repels like* — two positives push apart. For gravity, *like attracts like* — two surpluses pull together. Gravity is Coulomb with the sign flipped.
- **a strength.** Each has its own constant, and gravity's is fantastically smaller.

That one flipped sign is what separates the two worlds. Because like charges repel, matter neutralises itself into pairs and electricity *cancels out*, staying local and small — which is why you feel no electric force from a distant star. Because like masses attract, gravity does the opposite: it *accumulates*, never cancelling, growing without limit until it is the only thing that matters across a galaxy. The split between the small world and the large one is not an extra assumption in this picture. It is that sign.

It is worth being careful about what this does and does not mean. It is a resemblance noticed after the fact, not a foundation the picture was built on — the two forces were introduced here for entirely separate reasons, one to make matter and one to gather it, and neither was derived from the other. It does not say that gravity is *really* electricity in disguise, and it does not fold the two strengths into one number. Whether they are at bottom the same force is an open question, and the resemblance is a reason to ask it rather than an answer.

A universe you could almost draw

That is the appeal. In place of an unrepeatable explosion followed by a cascade of unseen ingredients, here is a universe you could nearly sketch on a napkin: a faint ripple in the electric field, curving sharply enough to make charge in units of exactly one and in two sizes — small dense protons, large diffuse electrons, and with the two sizes, the two masses; Coulomb assembling them into nuclei, atoms and chemistry; and Newton, acting on the energy of all that matter — on its surpluses and shortfalls, not on its bulk — gathering it into a cosmic web of filaments and clusters threaded by voids that push outward as they empty. One ripple, two old laws, three numbers.

It should be said just as plainly what this costs. The picture does not explain the accelerating expansion that dark energy was invented for — it says the universe coasts instead, which is a different claim and a checkable one. It does not explain why galaxies spin as fast at their edges as at their centres, the observation that dark matter was invented for; that problem is left exactly where Newton left it. And it does not yet explain why a nucleus is as small as it is, missing by a factor of about thirty. Three debts, stated openly.

So the simplest question of all is left standing: is nature really this economical? It may not be. But it is worth knowing what the simple answer looks like — and precisely where it fails — before reaching for the complicated one.

For the mathematics — the two-valued Poisson equations, the RealQM / RealNucleus computations, the cosmic web of overdense filaments and underdense voids, and the quantitative claims and open problems only gestured at here — see the companion technical article, C. Johnson, *Coulomb + Newton: Towards a Cosmology, and the treatise Real Quantum Mechanics (∼340 pp), both free from the RealQM gallery,* `https://claes542.github.io/RealMolecule/`.

For the blackbody argument — that a perfect Planck spectrum is the signature of a resonant lattice, not of a hot dense gas — see C. Johnson, *Computational Blackbody Radiation (online book).*